*.wet

Purpose

The wetland group has two input files declared in `type input_res` (`src/input_file_module.f90`):

- `wetland.wet` is the wetland object table. One record per wetland, naming the initial-condition set, hydrology profile, release rule, sediment profile, nutrient profile, and weir profile that apply.
- `hydrology.wet` is the wetland hydrology database. One record per named hydrology profile. The geometry is HRU-relative (fraction of HRU area, depth in mm) rather than absolute area and volume as for reservoirs.

Wetlands reuse the reservoir sediment, nutrient, weir, and initial-condition databases. The wetland records in `wetland.wet` point at the same `sediment.res`, `nutrients.res`, `weir.res`, and `initial.res` tables described in reservoir.res.

Source

- Readers: src/wet_read.f90, src/wet_read_hyd.f90
- Type definitions: src/reservoir_data_module.f90, `type reservoir_data_char_input` (shared with reservoirs) and `type wetland_hyd_data`
- Filename declarations: src/input_file_module.f90, `type input_res`

wetland.wet

Header lines: 2. The reader skips a title line and a column-header line, then reads one record per wetland with list-directed I/O. The first integer on each record is the wetland id (used as the index into `wet_dat_c`). The remaining columns are read into `type reservoir_data_char_input` in declaration order (the same type used by reservoirs).

#	Field	Type	Description
1	id	int	wetland id. Used as the array index into `wet_dat_c`
2	name	char(25)	wetland name
3	init	char(25)	initial-state pointer. Cross-walks with `init` in `initial.res`
4	hyd	char(25)	hydrology pointer. Cross-walks with `name` in `hydrology.wet`
5	release	char(25)	release rule pointer (decision-table name)
6	sed	char(25)	sediment pointer. Cross-walks with `name` in `sediment.res`
7	nut	char(25)	nutrient pointer. Cross-walks with `name` in `nutrients.res`

The example dataset writes a trailing `WEIR` column. The base type has six character fields (name, init, hyd, release, sed, nut). A weir pointer is held in the extension type `reservoir_data` (`character (len=25) :: weir`). The list-directed reader for `wetland.wet` does not consume a separate weir token into the base record; the source links wetland weirs through dedicated reservoir-conditions input. See the Important note below.

Example

`refdata/Osu_1hru/wetland.wet`:

```
wetland.       wet: written by SWAT+ edi#############for SWAT+ rev.60.5.4
     id                name        init         hyd         rel         sed         nut        WEIR
```

```
1          paddy0001    initwet2       paddy        weir     sedwet1     nutwet1       weir1
```

Notes:

- The title line in this dataset is partially corrupted ("wetland. wet:"). The reader skips it as a single text line, so the content does not matter as long as it stays on one line.
- The `rel` column value `weir` is matched against decision-table names. If no match exists in `res_rel.dtl` (or in the `ctbl_` conditional tables), the wetland will be flagged as missing a release rule.

hydrology.wet

Header lines: 2. The reader skips a title line and a column-header line, then reads one record per hydrology profile with list-directed I/O. Records are read into `wet_hyddb` (an array of `type wetland_hyd_data`) in declaration order.

#	Field	Type	Units	Default	Description
1	name	char(25)	none	“default”	hydrology profile name
2	psa	real	fraction	0.0	fraction of HRU area at the principal spillway. Surface-inlet riser flow starts at this level
3	pdep	real	mm	0.0	average water depth at the principal spillway
4	esa	real	fraction	0.0	fraction of HRU area at the emergency spillway. Overflow into the ditch starts at this level
5	edep	real	mm	0.0	average water depth at the emergency spillway
6	k	real	mm/hr	0.01	hydraulic conductivity of the wetland bottom
7	evrsv	real	none	0.7	wetland evaporation coefficient
8	acoef	real	none	1.0	volume-surface area coefficient for HRU impoundment
9	bcoef	real	none	1.0	volume-depth coefficient for HRU impoundment
10	ccoef	real	none	1.0	volume-depth coefficient for HRU impoundment
11	frac	real	none	0.5	fraction of HRU that drains into the impoundment

The reader fills defaults when fields are zero or non-positive:

- If `psa <= 0`, sets `psa = 0.08 * wet_hyd(ires)%pdep` (uses the previously initialized `wet_hyd`, not the just-read `wet_hyddb`).
- If `esa <= 0`, sets `esa = 1.5 * wet_hyd(ires)%psa`.
- If `evrsv <= 0`, sets `evrsv = 0.6`.

Example

`refdata/Osu_1hru/hydrology.wet`:

```
hydrology.wet: written by SWAT+ editor v2.2.0 on 2023-03-22 04:25 for SWAT+ rev.60.5.4
name                    hru_ps         dp_ps        hru_es         dp_es             k          evap    vol
paddy                     1.00        150.00          1.00        150.00           0.5          0.75
```

Editor headers `hru_ps`, `dp_ps`, `hru_es`, `dp_es`, `evap`, `vol_area_co`, `vol_dp_a`, `vol_dp_b`, `hru_frac` correspond to source fields `psa`, `pdep`, `esa`, `edep`, `evrsv`, `acoef`, `bcoef`, `ccoef`, `frac`. Column order is what matters.

Related files

- `reservoir.res` and `initial.res`. Shared tables. `wetland.wet` records point at `initial.res` names, and the sediment, nutrient, and weir lookups use the same databases as reservoirs.
- HRU connectivity. Each wetland is attached to an HRU; the geometry in `hydrology.wet` is expressed as fractions of HRU area.